

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I _Furthose samreen B (192511164) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student : *Furthose samreen B (192511164)*

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}$, $\text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 – Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbols

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Flu	Patient has Flu
Mea	Patient has Measles
Pne	Patient has Pneumonia

Rules

Rule 1

$$(F \wedge C) \rightarrow Flu$$

Rule 2

$$(F \wedge R) \rightarrow Mea$$

Rule 3

$$(C \wedge B) \rightarrow Pne$$

Facts

F
C
B

(b) Forward Chaining

Initial Facts

Fever
Cough
Breathlessness

Step 1

Check Rule 1
 $Fever \wedge Cough \rightarrow Flu$
Both are True.
Fire Rule 1.
Derived:
Possible Flu

Step 2

Check Rule 2
 $Fever \wedge Rash \rightarrow Measles$
Rash is False (not given).
Rule cannot fire.

Step 3

Check Rule 3

$\text{Cough} \wedge \text{Breathlessness} \rightarrow \text{Pneumonia}$

Both True.

Fire Rule 3.

Derived

Possible Pneumonia

Final Diagnoses

Patient A has Flu.

Patient A has Pneumonia.

(c) Backward Chaining

Goal

Pneumonia

Look for rule producing Pneumonia.

$\text{Cough} \wedge \text{Breathlessness} \rightarrow \text{Pneumonia}$

Sub-goals become

Need Cough

Need Breathlessness

Facts

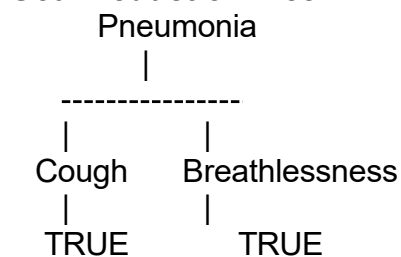
Cough ✓

Breathlessness ✓

Therefore

Pneumonia proved.

Goal Reduction Tree



Hence Goal Achieved.

CASE STUDY 2 – Autonomous Traffic Management Agent

(a) Unification

Rule 1

$$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$$

Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Step 1

Match

$\text{Vehicle}(x)$

$\text{Vehicle}(\text{Ambulance})$

Substitution

$$\theta_1 = \{x/Ambulance\}$$

Step 2

Apply substitution

EmergencyType(Ambulance)

Matches fact.

Final MGU

$$\theta = \{x/Ambulance\}$$

Therefore

ClearPath(Ambulance)

is inferred.

(b) Resolution Proof

Goal

Proceed(CarA)

Negate Goal

$\neg$ Proceed(CarA)

Convert to CNF

Rule 1

$\neg$ Vehicle(x) $\vee$ $\neg$ EmergencyType(x) $\vee$ ClearPath(x)

Rule 2

$\neg$ ClearPath(x) $\vee$ GreenSignal(x)

$\neg$ ClearPath(x) $\vee$ $\neg$ RedSignal(x)

Rule 3

$\neg$ Vehicle(x)

$\vee$ $\neg$ Behind(x,y)

$\vee$ $\neg$ GreenSignal(y)

$\vee$ Proceed(x)

Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)

Negated Goal

$\neg$ Proceed(CarA)

Resolution Steps

Vehicle(Ambulance)

•

Rule1

↓

ClearPath(Ambulance)

↓

Resolve with Rule2

↓

GreenSignal(Ambulance)

↓

Use

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

with Rule3

↓

Proceed(CarA)

Resolve

Proceed(CarA)

$\neg$ Proceed(CarA)

↓

□ (Empty Clause)

Hence proved.

(c) Analysis

Current KB only supports **one emergency vehicle**.

If there are two ambulances,

Example

Ambulance1

Ambulance2

the KB does not specify

- which vehicle has priority
- conflict resolution
- simultaneous green signals
- route coordination

Additional Rules

Priority(x) $\wedge$ Priority(y)

→ HigherPriority(x,y)

HigherPriority(x,y)

→ GreenSignal(x)

GreenSignal(x)

→ HoldSignal(y)

DifferentRoute(x,y)

→ AllowBoth

These rules prevent signal conflicts and ensure safe routing.

CASE STUDY 3 – Agricultural Expert System

(a) Convert into CNF

P1

Original

SoilDry → IrrigationNeeded

Remove implication

$\neg$ SoilDry $\vee$ IrrigationNeeded

Already CNF.

P2

IrrigationNeeded $\wedge$ CropWheat

→ ApplyDripMethod

Remove implication

$\neg(I \wedge C) \vee A$

DeMorgan

$\neg I \vee \neg C \vee A$

CNF obtained.

P3

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$

$\rightarrow \text{CropAtRisk}$

Remove implication

$\neg(\neg A \wedge C) \vee R$

DeMorgan

$A \vee \neg C \vee R$

CNF.

Facts

SoilDry

CropWheat

Final CNF

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

SoilDry

CropWheat

(b) Resolution Refutation

Goal

ApplyDripMethod

Negate Goal

$\neg \text{ApplyDripMethod}$

Resolution

SoilDry

•

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

↓

IrrigationNeeded

↓

Resolve with

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

↓

Using

CropWheat

↓

ApplyDripMethod

↓

Resolve
ApplyDripMethod

$\neg$ ApplyDripMethod

↓

□

Goal proved.

Resolution Tree

SoilDry

└─ IrrigationNeeded

CropWheat----- ApplyDripMethod

└─ $\neg$ ApplyDripMethod

□

(c) Remove SoilDry

Remaining Facts

CropWheat

Cannot derive

IrrigationNeeded

Therefore

ApplyDripMethod

cannot be proved.

Without

ApplyDripMethod

Rule 3 also cannot infer

CropAtRisk

Hence

CropAtRisk does NOT logically follow.

CASE STUDY 4 – Wumpus World

(a) Belief State

Percepts

Stench(1,2)

Breeze(1,1)

Glitter(2,2)

Rules

Stench $\rightarrow$ Adjacent Wumpus

Breeze $\rightarrow$ Adjacent Pit

Glitter $\rightarrow$ Gold

$\neg$ Wumpus $\wedge$ $\neg$ Pit $\rightarrow$ Safe

Using Modus Ponens

From

Stench(1,2)

Infer

Wumpus is adjacent to (1,2)

Possible cells

(1,1)

(2,2)

(1,3)

From

Breeze(1,1)

Infer

Pit adjacent to (1,1)

Possible

(1,2)

(2,1)

From

Glitter(2,2)

Infer

Gold at (2,2)

Derived Knowledge

- Wumpus adjacent to (1,2)
- Pit adjacent to (1,1)
- Gold at (2,2)

(b) Forward Chaining

Initial Facts

Stench(1,2)

Breeze(1,1)

Glitter(2,2)

Apply Rule 1

↓

Possible Wumpus

(1,1)

(2,2)

(1,3)

Apply Rule 2

↓

Possible Pit

(1,2)

(2,1)

Apply Rule 3

↓

Gold(2,2)

(c) Safety Analysis

Path

(1,1)

↓

(1,2)

↓

(2,2)

Analysis

- (1,2) may contain a Pit because of Breeze at (1,1).
- (2,2) may contain the Wumpus because of Stench at (1,2).
- Gold is located at (2,2), but the agent cannot prove that (2,2) is free of danger.
Since Rule 4 ($\neg \text{Wumpus} \wedge \neg \text{Pit} \rightarrow \text{Safe}$) cannot be app

Final Conclusion

The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is **not guaranteed to be safe**. The agent should gather additional percepts or explore alternative safe cells before attempting to collect the gold.